

Set of IP Addresses and Their Binary Representation

Index	Data	Index	Data
0	1.2.3.1	000	0001_0010_0011_0000
1	1.2.4.2	001	0001_0010_0011_0001
2	1.2.5.3	010	0001_0010_0011_0010
3	1.2.6.4	011	0001_0010_0011_0011
4	1.2.7.5	100	0001_0010_0011_0100

throughput, which is of utmost importance when we talk about the line speeds which are approaching 10 Gbps.

Software and hardware CAM

Hardware search engines are faster than algorithmic approaches for search-intensive

applications. Also, the turn-around time of a hardware CAM would be highly deterministic, while that of software would depend on how efficiently the search-tree balancing algorithm is implemented.

However, software

itself has a priority encoder that helps in resolving the winning address.

Example of a table look-up in a CAM

Consider the accompanying table to understand the look-up functionality. This table has five entries, each representing an IP address. Now if a datagram was to be forwarded from address '1.2.3.1' to '1.2.4.2', the IP packet would have the source IP address as '1.2.3.1' and '1.2.4.2' would be the destination address.

Taking the above two addresses as communicating hosts, we can think of multiple scenarios like:

1. Both the hosts reside in the same network
2. Hosts reside in the neighbouring networks
3. Hosts are statically located in two distant networks
4. While one host is statically located in a network, the other host is mobile

Both the hosts reside in the same network. Once a router gets a packet, the first task that the router has to perform is to know whether the destination address of the packet is within its network or outside. If the host with the address of '1.2.4.2' was within the same network, the router need not do anything as the hub or the switch sitting within that network would take care of delivering that packet to the desired destination. In fact, if a switch sits in the local network, it would not even forward the packet to the router if the packet belongs to a local host.

It should be noted that if a switch was used in the network it would consider the MAC address of the hosts as the addresses, while if a hub was used all the data would be broadcast throughout the network and whichever host is the rightful recipient (based on the MAC address) would take the packet while all others would discard the information.

Hosts reside in the neighbouring networks. However, if the destination

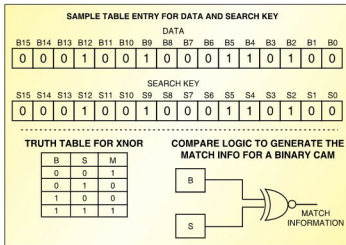


Fig. 1: Working of a binary CAM cell

the CAM and identifies the next hop the packet needs to take to eventually reach its destination.

The two major functions of a CAM include storage of information and look-up logic to find the data being searched for. Looking at these tasks a software geek may say that he would store the data in a regular memory and then have software sitting around it to do the look-up instead of using a special hardware. This is where we try to define the two ways of building a CAM:

1. Build a software wrapper around the memory (SRAM/DRAM) to perform the look-up functions for the searched data

2. Build a hardware that stores data (functions as a memory) as well as performs the look-up

Both the approaches have their pros and cons. In the second approach, the entire functionality is carried out in the hardware. This, as observed practically, dramatically improves the

would result in lower costs.

Hybrid CAMs integrate the robustness of a hardware CAM while providing the flexibility of a high degree of software programming.

Types of CAMs

Fundamentally, CAMs are of two types: binary and ternary. As the name indicates, a binary CAM has two bit states ('1' and a '0'), while a ternary CAM has '1', '0' and a third state called "don't care." So a search-hit on a binary CAM would be an exact match search, while a search-hit on a ternary CAM could be an exact match or a partial match.

Currently, ternary CAMs are *de facto* in the industry. These bring the possibility of multiple match in a look-up table. All the access list (ACL) tables for packet forwarding use a popular address resolution algorithm called the longest prefix match where the addresses are placed in the CAM based on certain priorities and the CAM by